2022

COMPUTER SCIENCE AND ENGINEERING

Paper : CSEL-1016

(Real Time Systems)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question no.*** **1,** ***question no.*** **2** and ***any four*** from the rest.

1. Answer ***any five*** questions : 2×5

(a) Consider a system of three independent preemptable periodic tasks :

T1 = (8, 1), T2 = (4, 3) and T3 = (12, 1)

All jobs have phases equal to zero and relative deadlines are equal to their periods. Can this system be scheduled using the 'earliest deadline first' algorithm? Explain your answer.

(b) Can the above system be scheduled using the monotonic rate algorithm? Explain your answer.

(c) Explain what priority inversion is. When does it occur?

(d) Explain why the basic priority ceiling protocol is immune to deadlock.

(e) Justify which scheduling policies are most suited for controlling a set of periodic tasks.

(i) FCFS

(ii) Least laxity first

(iii) Earliest deadline first

(iv) Rate monotonic policy schedule.

(f) Scheduling of tasks is a critical consideration in RTOS. Which of the following best describes the scheduling policy design? Justify your answer.

(i) The scheduler must follow a pre-emptive policy

(ii) The scheduler must not use a pre-emptive policy option

(iii) The scheduler must not only use pre-emptive policy options with priority considerations

(iv) The scheduler must not use a pre-emptive policy option but must employ priority consideration.

(g) If Priority Inheritance Protocol is used as a semaphore protocol in a real-time system, then the system is guaranteed to be deadlock-free. Justify.

2. Answer the following questions (*any five*) : 4×5

(a) Critically Comment on the desirable characteristics of a real-time system.

(b) What is a rate-monotonic scheduling algorithm? What are the various assumptions in this algorithm? Explain the rate-monotonic algorithm with a suitable example.

(c) Discuss the relative advantages and disadvantages of clock-driven and priority-driven algorithms. Explaining when it is appropriate to use a clock-driven scheduler and when one should use a priority-driven one.

(d) What do you mean by the scheduling point of a task scheduling algorithm? How are scheduling points determined in clock-driven, event-driven, and hybrid schedulers?

(e) Explain how does a real-time database differ from a conventional database.

(f) Compare and contrast basic priority inheritance and priority ceiling protocols.

3. (a) Find the RM, EDF, LST and FIFO schedule for the set of periodic tasks T1 = (2, 1) and T2 = (5, 2.5).

(b) How will you find the critical instances for a given set of real-time tasks? Make a suitable set of assumptions.

(c) What are proficient and optimal schedulers? 5+3+2

4. (a) Comment on the following :

(i) Job slices' for a given set of periodic tasks.

(ii) Effective release time and effective deadlines.

(b) Find the job schedule using the basic priority inheritance protocol for the following set of jobs with associated parameters. 3+7

Job	Release Time	Execution Time	Priority	Critical Section
J1	7	3	1	[Shaded;1]
J2	5	3	2	[Black;1]
J3	4	2	3	
J4	2	6	4	[Shaded;2.5 [Black;1.5]]
J5	0	6	5	[Black;4]

5. The periodic tasks (3, 1), (4, 2) and (6,1) are scheduled according to the rate-monotonic algorithm

(a) Draw the time-demand functions of the tasks with justifications.

(b) Are the tasks schedulable? Justify your answer.

(c) Can this graph be used to determine whether the tasks are schedulable according to an arbitrary priority-driven algorithm? Explain your answer. 5+3+2

6. (a) Critically comment on the followings :

   (i) Constraints to determine the frame size for a given set of periodic tasks

   (ii) Tick Scheduling

   (iii) Controller Area Network

   (iv) Time Demand Analysis.

   (b) Safety-critical real-time systems do not have a fail-safe state. – Critically comment. 8+2

7. (a) "The traditional 2PL protocol is not suitable for real-time database." – Critically comment.

   (b) Generate examples of an application where it is permissible to relax one or more ACID properties of transactions in real-time database systems.

   (c) Briefly explain the different communication models in IoT. 2+4+4